Argha Kundu

Kolkata, West Bengal — arghakundu9999@gmail.com — +91 9330713093 — linkedin.com/in/arghakundu — github.com/Meexargha

Professional Summary

Backend-focused Software Engineer with hands-on experience developing REST APIs using Node.js, Express.js, and MongoDB. Experienced in JWT authentication, authorization, RBAC, database indexing, query optimization, atomic database operations, and modular backend architecture. Strong foundation in Data Structures & Algorithms, OOP, SQL, and software development practices.

Education

Academy of Technology Hooghly, West Bengal
Bachelor of Technology in Electronics and Communication Engineering *2022 – 2026*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management, SQL

Technical Skills

Languages: C++, Python, JavaScript
Backend: Node.js, Express.js, REST APIs
Databases: MongoDB, Mongoose, PostgreSQL
Security: JWT, Authentication, Authorization, RBAC, Input Validation
Concepts: Data Structures & Algorithms, OOP, SDLC, Database Indexing, Query Optimization
Tools: Git, GitHub, Postman, Vercel, Cloudinary

Projects

Streamify – Video Sharing REST API — GitHub

- **Technologies:** Node.js, Express.js, MongoDB, Mongoose, JWT, Cloudinary, Postman
- Built a modular layered REST API with JWT Bearer authentication and ownership-based authorization to protect video operations and restrict resource modifications to authenticated owners.
- Integrated Cloudinary for video and thumbnail storage while maintaining media URLs and public IDs in MongoDB, separating media storage from application data.
- Optimized MongoDB query paths using timestamp and compound indexes, validated execution plans with `explain()` and index scans, and used atomic `$addToSet` and `$pull` operations for like and view relationships.

Hotel Staff Management Backend System — GitHub

- **Technologies:** Node.js, Express.js, MongoDB, Mongoose, JWT, RBAC, REST API
- Developed a backend database solution to automate hotel staff record handling using structured Mongoose schemas, validation, and REST APIs.
- Implemented JWT authentication and RBAC middleware to secure backend endpoints and control access based on staff roles.
- Integrated a currency conversion API to support currency-related operations through backend service endpoints.

Reconfigurable Absorber/Reflector Metasurface for 6G Communication *Final Year Project*

- Developed reconfigurable absorber/reflector structures for high-frequency communication systems using electromagnetic simulation.
- Evaluated electromagnetic absorption and reflection characteristics of the designed structures across targeted operating frequencies.
- Studied structural tuning approaches to achieve dynamically switchable absorber and reflector behavior for next-generation communication systems.

Achievements

- Solved **300+ problems on LeetCode**, strengthening Data Structures & Algorithms and problem-solving skills.
- Secured **Global Rank 3929** in TCS CodeVita 2025.
- Ranked **1st among 371 participants** on Academy of Technology leaderboard (July 2025).
- Selected as a **Finalist in a college hackathon** and contributed to the college technical community through the tech team and technical fests.